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A Final Year Synopsis Presentation
on

TrustTag: A Lost Item Finder App Using QR Code and Location Tracking

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Introduction

In today's fast-paced world, losing valuable items is a common and stressful experience.

Introducing TrustTag, a comprehensive mobile application designed to bridge the gap between those who lose items and those who find them.

Our solution empowers users with a smart, QR Code based identity system for their belongings.

Beyond lost items tracking TrustTag also integrates SOS Alert and Locating Loved Ones features, making it an all-in-one personal safety and recovery platform.

Objective

Here are the core development objectives that will guide the building of our app.

Objective 1 : To Develop a Secure and Scalable User Authentication & Data Management System

The foundation of the app is user trust. This objective focuses on building a robust backend that securely handles user registration, login, and profile management. It must ensure that personal data is encrypted and stored safely, with clear segregation between private account details ant the user- chosen shareable details for the QR Code.

Objective 2: To Engineer a Dynamic QR Code Generation & Management Engine

This is the core technical innovation. The objective is to build a system that can generate a single, persistent QR code for each user, whose displayed information can change dynamically based on the item's status (active if lost, and deactivate if found). The engine must also support extensive user customization for size, style, and embedding images, without altering the underlying data link.

Objective 3: To Implement a Real-Time Notification and Geolocation Tracking System

This objective focuses on creating the alert mechanism that provides the primary value to a user who has lost an item. It involves Push Notifications for the app, SMS Gateway to send instant alerts. Crucially, it must capture and relay the geolocation coordinates from scan event to pinpoint where the item was found, enabling a quicker recovery.

Problem Statement

1. **Inefficient Traditional Methods**
   Manual methods like notice boards, announcements, social media are slow, unsecure, and depend on goodwill.

2. **Eco-System Locked Tech Stack**
   Modern tools like Apple's Find My or Samsung SmartThings are effective but only work within their own brand ecosystems, excluding a vast majority of users and creating a fragmented market.

3. **Hardware Dependency**
   Hardware trackers like Tile, AirTag, BLE/GPS devices are costly, require maintenance, and have limited range.

4. **Privacy & Security Issues**
   Existing BLE/GPS systems store user data on centralized servers, risking misuse and unauthorized tracking. Lack user control over shared information. Users are hesitant to share personal data publicly.

5. **Connectivity Challenges**
   There is no secure, real-time communication channel that connects a finder and an owner instantly, without compromising personal data, and that works for everyone regardless of their device.

Modules

User Authentication and Registration Module

Purpose: To manage user registration, authentication, and profile management securely.

Functionality: When a user registers, their details are verified using OTP through Firebase. A unique userID is generated and used for linking items and QR codes to that account.

QR Code Generation Module

Purpose: To create unique, encrypted, and customizable QR codes for each registered item.

Functionality: Users register items, and the system generates a secure tokenized QR code that can be printed and attached to belongings.

Lost Item Management Module

Purpose: To handle item status updates (lost/ recovered) and manage related data in real time.

Functionality: When an item is reported lost, the QR token is activated. If scanned, it shows only limited owner details, preserving privacy. Once recovered, the QR automatically deactivates.

Scan and Notification Module

Purpose: To connect finders and owners in real time through secure, automated notifications.

Functionality: When a finder scans an active QR, the backend authenticates the request, retrieves owner-approved info, and triggers multi-channel notifications with location details.

SOS Alert Module

Purpose: To ensure user safety by providing an emergency alert system within the app.

Functionality: The SOS module uses GPS and Firebase Cloud Messaging to transmit real-time coordinates to emergency contacts during distress situations.

Loved One Locator Module

Purpose: To enhance personal connectivity and location sharing among trusted users.

Functionality: This module allows family members or trusted contacts to view the user's live location securely, ensuring safety and transparency in movement tracking.

Feasibility Study

1. RFID & NFC Systems
- Need special readers; short range; high setup cost.
- Suitable for institutions, not for personal lost items.

2. BLE Trackers (AirTag, Tile, SmartThings)
- Short-range + low energy.
- Ecosystem-locked, requires hardware & batteries.
- Privacy concerns; not universal.

3. GPS & IoT Tracking
- Accurate but expensive, power-heavy, requires devices & data plans.
- Complex for daily personal use.

4. QR-Code-Based Solutions
- Low-cost & universal(any smartphone camera can scan).
- Existing systems are static, lack real-time activation/deactivation & privacy control.

5. Security & Privacy Research
- Centralized trackers risk data misuse and unauthorized tracking.
- Need encryption + selective data sharing.

6. Research Gap Identified
- No existing solution that is: Hardware-free Cross-platform, Real-time, Low-cost, and Privacy-controlled.

7. Proposed Work
- Lost Item Finder App uses:
  - ✔ Dynamic QR codes
  - ✔ Cloud synchronization
  - ✔ GPS-based alerts
  - ✔ Strong privacy controls
- Fills gap by offering a universal, scalable, and secure lost-item recovery system.

Workflow

1. User Registration →

User signs up using Name, Email, Mobile, WhatsApp number. OTP verification through Firebase. Profile and privacy preferences saved in cloud.

2. Item Registration →

User adds item details (name, description, optional photo). System generates unique, encrypted QR code. User customizes and prints QR to attach to belongings

3. QR Activation (Mark Item Lost) →

When item is lost → user taps “Mark Item Lost”. QR status changes to Active in cloud database. Ready for scanning by finder.

4. Finder Scans QR Code →

Finder scans using phone camera → opens secure webpage. Only limited, user-selected details are shown (no full data exposed).

5. Real-Time Notifications →

On scan, the owner instantly receives: In-app notification, SMS alert, and GPS location of scan

6. Item Recovery →

Owner contacts finder → retrieves item. Taps “Item Recovered”. QR automatically deactivates (privacy restored).

7. Additional Safety Features

SOS Alert: Sends emergency message + live location to trusted contacts.
Loved Ones Locator: Consent-based real-time location sharing.

Technology Stack

- **Frontend:** Android (Kotlin) + React Web
- **Backend:** Python (FastAPI/Flask)
- **Realtime & Auth:** Firebase Auth + Firestore
- **Notifications:** FCM + Twilio
- **Maps:** Google Maps API
- **QR:** qrcode/segno (Python) → CDN (Firebase Storage / S3)
- **Security:** HTTPS, JWT, SHA-256, AES-256
- **Hosting & CI/CD:** Cloud Run / Render + GitHub Actions
- **Monitoring:** Sentry + Firebase Analytics

Requirement Analysis

Functional Requirements

1. User registration & login with OTP
2. Profile management (view/edit, privacy preferences)
3. Item registration
4. QR code generation
5. Mark Item Lost (activate QR in cloud)
6. Mark Item Recovered (deactivate QR)
7. Public scan endpoint → show owner-approved limited info
8. Real-time notifications to owner
9. SOS alert — single-tap send live location to contacts
10. Loved Ones Locator — consent-based live location sharing

Non-Functional Requirements

1. Performance: scan-to-notify latency < 3 seconds (target)
2. Availability: ≥ 99% uptime
3. Scalability: support large amount of concurrent users
4. Security
5. Privacy: selective disclosure of owner fields
6. Reliability: retry logic for failed notifications
7. Usability: intuitive UI; minimal steps for core flows
8. Maintainability

Hardware Requirements

1. User device: any smartphone with camera, GPS & internet
2. Server: cloud hosting (e.g., Firebase + container host)
3. No external hardware needed (no RFID/BLE/GPS trackers)

Software Requirements

1. Frontend: Android app (Kotlin) — Camera & Maps integration
2. Backend: Python (Flask/FastAPI) with RESTful APIs
3. Database: Firebase Realtime DB / Firestore (primary)
4. Notifications: Firebase Cloud Messaging (FCM)
5. SMS: Twilio (or provider) integration
6. Maps & Geocoding: Google Maps API
7. QR generation libraries (qrcode/segno) & image CDN

Expected Outcomes

1. Intelligent QR-Based Identity for Everyday Items
A new way of tagging belongings using dynamic, secure QR codes, allowing any lost item to communicate with its rightful owner without specialized hardware.

2. Zero-Hardware Smart Tracking System
Introduces a completely hardware-free tracking solution – no RFID tags, no BLE devices, no GPS trackers – making it universally accessible and eco-friendly.

3. Privacy-First Lost & Found Ecosystem
Owner maintains complete control over which details are shown. The system introduces the concept of “conditional visibility”, where data appears only when the item is reported lost.

4. Instant Global Recovery Network
Since QR can be scanned by any smartphone anywhere, the system creates a global recovery network without requiring people to install the app.

5. Automated Owner-Finder Communication Without Number Sharing
Breakthrough in privacy: Finder and owner connect without revealing personal contact numbers, enabled through controlled access and cloud-based routing.

6. Integrated Safety Architecture
The app not only recovers lost items but also enhances personal safety through SOS alerts and Loved Ones Location Tracking — combining two domains into one platform.

7. Dynamic QR Lifecycle Management
First-of-its-kind feature where QR codes: Activate when item is lost, Deactivate when found, Preserve privacy during inactive stages.

8. Institutional-Ready Lost & Found Automation
The solution can transform campuses, airports, malls, and offices into smart lost-and-found ecosystems, eliminating manual dependency.

Conclusion

- ✔ The Lost Item Finder App provides a smart, secure, and affordable solution for recovering lost belongings. By integrating dynamic QR codes, GPS tracking, and cloud-based notifications, the system overcomes the limitations of traditional lost-and-found methods and hardware-dependent tracking technologies.
- ✔ The app ensures complete user privacy and real-time communication, allowing owners and finders to connect instantly without exposing personal details.
- ✔ With features like SOS Alerts and Loved Ones Locator, the application extends beyond item recovery and contributes to personal safety and digital well-being.
- ✔ Overall, the project delivers a universal, scalable, and user-friendly platform that modernizes the lost-item recovery process and enhances everyday convenience for individuals and institutions.

Project Development Schedule

September | October | November | January | February | March

Planning

Requirement Gathering, Problem Understanding, Literature Review, Feasibility Study, Finalize SRS, Prepare Use Cases, Identify System Modules, Technology Stack Finalization

Design

Design Approach, High-Level Architecture Design, Prepare UML Diagrams, Detailed System Architecture, Database Design (Firestore Schema), API Structure

Development

Frontend Mobile Application Setup UI Layouts, Splash & Login Screens, Backend Setup, Authentication Module, User Registration + Login Functionality, QR Code Generation Module, Tokenization Logic (SHA-256), Customization Features, Lost Item Activation Module, Finder Interface, Limited Data Display Logic, Notification Engine, GPS Location Capture on Scan, SOS Module + Loved Ones Locator Module, Location Sharing Workflows

Testing

Testing (Unit, Integration, Functional), Performance Evaluation, Bug Fixes

Deployment

Final Deployment, Documentation, Report Preparation, Demo Presentation

Thank you!